

Facade Design Pattern

Simplifying Complex Systems

Md Mehedi Hasan Maruf-2003037 Md Shahidul Islam Dipu-2003040
Eazdan Mostafa Rafin-2003055

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Department of Computer Science and Engineering, RUET

What is a Design Pattern?

- Reusable solutions to common software design problems
- Not code, but templates for solving problems
- Enhance code maintainability and scalability

Types of Design Patterns:

- **Creational Patterns**
- **Structural Patterns**
- **Behavioral Patterns**

Facade Design Pattern

- **Type:** Structural Design Pattern
- **Objective:**
 - Provide a simplified interface to complex subsystems.
 - Reduce dependencies between the client and subsystem classes.

When to Use Facade

- Complex Subsystems with many interdependent classes
- Need to provide a simple interface for clients
- Layered systems requiring interaction with lower layers
- Integrating with legacy or third-party code

Key Components

1. **Facade:** Simplified interface for the client
2. **Subsystem Classes:** Complex classes handling specific functionalities
3. **Client:** Interacts only with the Facade

C++ Code Example

```
1 // Facade Design Pattern in C++
2 #include <iostream>
3 using namespace std;
4
5 class Light {
6 public:
7     void dim() { cout << "Lights dimmed.\n"; }
8 };
9
10 class MusicSystem {
11 public:
12     void play() { cout << "Playing music.\n"; }
13 };
14
15 class HomeTheaterFacade {
16     Light light;
17     MusicSystem music;
18 public:
19     void activateMovieMode() {
```

- Simplifies interactions with complex systems
- Promotes loose coupling between client and subsystem
- Enhances code readability and maintainability
- Facilitates easier testing and debugging

- **Hides Too Much:** Can block access to advanced features.
- **Overdependence:** Makes it harder to work without the Facade.
- **Maintenance Overhead:** Needs regular updates to stay in sync.

- **Home Theater Systems:** One button dims lights, starts the projector, and plays your movie.
- **Computer Start-Up:** Turns chaos into a smooth boot-up process.
- **Online Shopping:** Handles payments, inventory, and shipping in the background.

The Facade Design Pattern simplifies complex systems by providing a clean and easy-to-use interface, making your code more readable, maintainable, and less dependent on the intricate details of subsystems.